

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

Priya is analyzing encrypted messages in a research project. She wants to analyze the frequency of each character in a given paragraph. The characters should be stored in a TreeMap so that the output is sorted in ascending order of characters automatically.

You are required to build a Java program that:

Uses a `TreeMap<Character, Integer>` to count how many times each character appears in the message. Ignores spaces and considers only alphabets (case-sensitive). Outputs the frequencies of characters in sorted order.

You must use a TreeMap in the class named MessageAnalyzer.

Input Format

The first line of input contains an integer n, the number of lines in the message.

The next n lines each contain a string (the encrypted message line).

Output Format

The first line of output prints: "Character Frequency:"

Then print each character and its frequency in the format: "<character>: <count>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2
Hello World
Java

Output: Character Frequency:

H: 1

J: 1

W: 1

a: 2

d: 1

e: 1

l: 3

o: 2

r: 1

v: 1

Answer

```
// You are using Java
import java.io.*;
import java.util.*;
```

```
class MessageAnalyzer{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        sc.nextLine();
```

```
        TreeMap<Character, Integer> msg=new TreeMap<>();
```

```
for(int i=0;i<n;i++){
    String s=sc.nextLine();

    for(char ch: s.toCharArray()){
        if(Character.isLetter(ch)){
            msg.put(ch, msg.getOrDefault(ch,0)+1);
        }
    }

}
System.out.println("Character Frequency:");
for(Map.Entry<Character,Integer> entry : msg.entrySet()){
    System.out.println(entry.getKey()+" "+entry.getValue());
}
}
```

Status : Correct

Marks : 10/10